

1. Local Filtering Algorithm

Step 1: Capture Video Frames

The camera continuously captures frames.

Step 2: Object Detection

Use a lightweight AI model to detect cars, trucks, motorcycles, buses, pedestrians

Step 3: Tracking

Assign IDs to moving objects across frames.

Step 4: Event Detection Rules

Trigger only important events such as: Accident/collision detected, Speed violation, Vehicle stopped too long, and Traffic congestion

Step 5: Local Summarization

For non-critical data, the camera creates periodic summaries such as vehicle count and average speed every minute instead of transmitting raw video:

```
Every 1 minute:  
vehicle_count = total vehicles  
avg_speed = average speed  
vehicle_type_count = {cars:20, trucks:4}
```

Pseudocode

```
while True:  
    frame = capture_frame()  
    objects = detect_objects(frame)  
    tracked_objects = track(objects)  
  
    for obj in tracked_objects:  
        if detect_accident(obj):  
            send_immediately(obj)  
  
        elif speed_violation(obj):  
            send_immediately(obj)  
  
        elif illegal_parking(obj, duration > 30):  
            send_immediately(obj)  
  
    update_local_summary(tracked_objects)  
  
    if end_of_minute():  
        send_summary()
```

2. Criteria for Interesting Data

The data can be divided into:

A) Immediate Transmission (Real-Time Alerts)

Sent instantly to central server.

Example:

```
{
  "timestamp": "10:05:00",
  "camera_id": "CAM-45",
  "event": "Accident",
  "location": "Sidi Gaber"
}
```

B) Batch Transmission (Midnight Reports)

Sent once daily.

Example:

```
{
  "date": "2026-04-27",
  "camera_id": "CAM-45",
  "total_vehicles": 18000,
  "avg_speed_day": 42
}
```

3. Average Traffic Density

We should calculate without storing full video.

Formula

Average Traffic Density:

Density = Total Detected Vehicles / Road Length

Sliding Window Method

Maintain only aggregated counters; no need to store full frames or raw video.

4. Partitioning Strategy

To ensure sightings of the same vehicle go to the same analytics node.

Option 1: Geographic Zone Partitioning

Partition by city zones (east, central, west)

Option 2: License Plate Hash Partitioning

Partition by license plates.

5. Out-of-Order Data Arrival

If Camera B's data arrives before Camera A's due to network delays, the server should process events using the event timestamp instead of arrival time.

Technologies such as Apache Kafka and Apache Flink support event-time processing and watermarks to handle delayed events.

6. Unique Vehicles Through the City Center

To count unique vehicles across all 5,000 cameras, the system can use license plate recognition, but the problem is large memory consumption

For large-scale counting, HyperLogLog is recommended because it estimates unique counts efficiently using less memory.

7. Handling Duplicate Events

Duplicate events can happen due to retries or multiple cameras capturing the same vehicle.

The system should generate a unique event ID and use time-window-based deduplication to ignore repeated events.